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Optimal skin regeneration after full thickness thermal burn injury in the spiny mouse, *Acomys cahirinus*

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ABSTRACT

The spiny mouse, *Acomys cahirinus*, shows remarkable regenerative abilities after excisional skin wounding by regrowing hair, sebaceous glands, smooth muscle, skeletal muscle and dermis without scarring. We have asked here whether this same regeneration can be seen after full thickness thermal burn injuries. Using a brass rod thermal injury model we show that in contrast to the lab mouse, *Mus musculus*, which forms a thick scar covered by a hairless epidermis, the spiny mouse regenerates all the tissues injured — skeletal muscle, dermis, hairs, sebaceous glands such that the skin is externally indistinguishable from its original appearance. Re-epithelialization is faster in *Acomys* than in *Mus* but ingression and proliferation of dermal fibroblasts is the same in both species. After 3 weeks the wound epithelium of *Acomys* has developed a covering of new hair follicles in contrast to *Mus*. The skeletal muscle of the panniculus carnosus in *Mus* shows some regeneration but it is incomplete and fibrotic whereas the *Acomys* muscle is replaced perfectly. There are differences in the macrophage profiles which invade the damaged tissues such as the absence of F4/80 or MOMA-2 +ve cells in *Acomys* which likely reflect different cytokine profiles resulting from the same injury in these two species.

1. Introduction

The hypertrophic scar which inevitably follows burn injury in mammals consists of a hairless, non-functional epidermis and a linear deposition of collagen in the dermis which lacks the flexibility of normal skin. This represents the greatest unmet challenge in burn recovery and in the human population produces severe problems of pain, stiffness, disabling joint contractures and poor quality of life [1,2].

The fibroblasts present in the hypertrophic scar generate excessive levels of collagen, smooth muscle actin, growth factors such as connective tissue growth factor, transforming growth factor-β pathway components and a variety of extracellular molecules such as osteopontin, biglycan and versican [3]. Although there are differences in the synthesis and cytokine profile of hypertrophic burn scars and normal scars generated by excisional wounding [1] they both suffer from the absence of hairs, dermal adnexa and abnormal matrix molecule deposition and, at least to this superficial degree, show considerable similarities.

Since we have recently discovered that a unique mammal, the spiny mouse of the genus *Acomys* can regenerate its skin perfectly following full thickness excisional wounding and replace hairs, sebaceous glands, erector pili smooth muscles, dermis, skeletal muscle of the panniculus carnosus [4–6] it was of obvious interest to determine whether this animal can do the same after burn injury to the skin. We describe here the response of *Acomys*

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cahirinus skin to full thickness thermal burn injury and compare it to the response of the CD-1 outbred laboratory mouse, *Mus musculus* as an example of non-regenerating, scarring mammalian skin. We show that in*Acomys* 6-8 weeks after injury there is no scar, the individual layers of the skin have regenerated and full replacement of hairs has taken place compared to *Mus* in which a hairless scar is generated. We compare the progress of scarring/regeneration in these two species in terms of histological appearance, cellular proliferation and macrophage profiles in order to begin an understanding of how the skin can regenerate after burn injury.

2. Material and methods

All animal procedures were approved by the Institutional Animal Care and Use Committee at the University of Florida. *A. cahirinus* were obtained from our breeding colony housed at University of Florida. Groups of animals, n=3 for each sample time, of mixed sex were used, aged from 6 months to one year old. *M. musculus* of the outbred CD-1 strain were purchased from Charles River and similarly sized mixed sex groups of animals used for the experiments of age 3 months to 6 months old.

Burns were administered identically to each animal by first anaesthetizing animals with iso-flurane and then shaving the dorsum. A brass rod heated to 100°C in boiling water was placed onto the mid-dorsal shaved area for 10s. The brass rods were 1.5cm in diameter and weighed 25g. Animals were each given 1ml of warmed sterile saline and 0.1mg/kg of buprenorphine before they had recovered from anesthesia. Buprenorphine administration was repeated at 12h intervals for 48h.

At regular time periods after injury, animals were terminally anaesthetized and the wounds fixed in 4% para-formaldehyde for 2h at 4°C for frozen sections or 10% neutral buffered formalin (NBF) for paraffin wax sections. After fixation the wounds for frozen sectioning were washed in phosphate buffered saline (PBS) pH 7.6 three times for 10min each and then placed in 30% sucrose made up in PBS overnight at 4°C. Wound tissues were then placed in disposable base moulds in OCT and frozen at −20°C. The following day 20μm sections were cut on a Leica cryostat and sections mounted on Superfrost plus slides.

The wounds for wax sections had the NBF washed out with PBS then they were dehydrated in graded ethanols (70%, 90%, 95%, 100%, 100%) 1h each, then xylene placed into the final 100% ethanol in 50:50 proportion, then 100% xylene and placed into the wax oven at 68°C. After 10min the samples were immersed in wax for 1h and the wax replaced 3× then the samples were embedded prior to sectioning. Paraffin wax sections were cut at 10μm, placed on Superfrost plus slides, dried overnight on a warm plate and used for haematoxylin and eosin staining and Mason's trichrome staining using standard techniques. They were also used for immunocytochemistry with a proliferating cell nuclear antigen (PCNA) antibody (Dako, clone PC10) with a Vector-Elite kit. Sections were dewaxed, endogenous peroxidases removed with hydrogen peroxide in methanol (0.3%), rehydrated in graded ethanols to PBS. A blocking solution of 1% goat serum in PBS was placed on the sections for 1h at room temperature and then sections were incubated with the PCNA antibody diluted 1 in 250 in blocking solution overnight at 4°C. After washing in PBS, biotinylated anti-mouse IgG at 1 in 200 dilution was placed on the sections for 1h at room temperature, the sections washed in PBS, then the ABC reagent (Vector-Elite PK-6100) applied for 1h followed by DAB reagent (Vector Labs peroxidase substrate kit, SK-41000) for 2min. The slides were then dehydrated and coverslips mounted with Cytoseal XYL.

Frozen sections were prepared at 20μm and used for immunocytochemistry with the following macrophage antibodies: F4/80 (Abcam ab6640), CD206 (Abcam, ab64693), CD86 (Abcam, ab53004), MOMA-2 (BioRad, MCA519G) and Iba-1 (Wako). For immunocytochemistry these cryostat sections were blocked in a mixture of 10% donkey serum, 0.5% Triton, 0.1% Tween and 0.1% fish skin gelatin for 1h, the antibodies were diluted 1 in 100 in blocking solution and incubated overnight at 4°C. After washing in PBS the slides were incubated at room temperature in the dark for 1h with the relevant Alexa Fluor 488 secondary antibody made in donkey at a dilution of 1 in 200 in blocking solution. Sections were mounted in 80% glycerol containing 5μg/ml Hoechst 33258 (Invitrogen).

Immunofluorescence was imaged on an Olympus IX81 inverted microscope allowing the camera to find the best exposure settings. Excitation and emission settings for visualizing the three standard flurophores were: DAPI Ex 350, Em 461; FITC Ex 475, Em 588; DSred Ex 558, Em 583.

In order to count percent proliferating (PCNA+ve) cells, the total number of nuclei in the field of view of the microscope at 40× magnification were counted as well as the number of positive cells in that same field of view. A percent +ve figure was thus obtained and then another adjacent field of view was similarly counted. 4 fields of view were counted on any one section and the same process repeated for a total of 3 sections. Approximately 1000 cells were counted for each estimation.

3. Results

3.1. Normal skin

The overall thickness of *Mus* and *Acomys* skin is approximately the same although the individual components vary somewhat in thickness (Fig. 1A and B). The epithelium of *Acomys* is thicker because it contains more cell layers, there is a thicker stratum corneum and the large spiny hairs have far larger follicles and penetrate far deeper into the dermis than in *Mus*. The dermis is thicker in *Mus* although the *Acomys* dermis has strands of connective tissue penetrating all the way to the panniculus carnosus. To compensate for the thicker *Mus* dermis, the adipocyte layer of the hypodermis is thicker in *Acomys* and the skeletal muscle of the panniculus carnosus (pc) is considerably thicker in *Acomys*. Below the panniculus carnosus in both species is the connective tissue fascia (Fig. 1A and B).

3.2. External aspects of burn injury

Following the burn injury with brass rods heated to 100°C the 1.5cm circle of skin blanches white and remains in place as a dried piece of tissue consisting of dead dermis, epidermis and densely packed hair follicles (Fig. 2A and B). Over the subsequent 2 weeks the *Mus* skin scab remains in place until

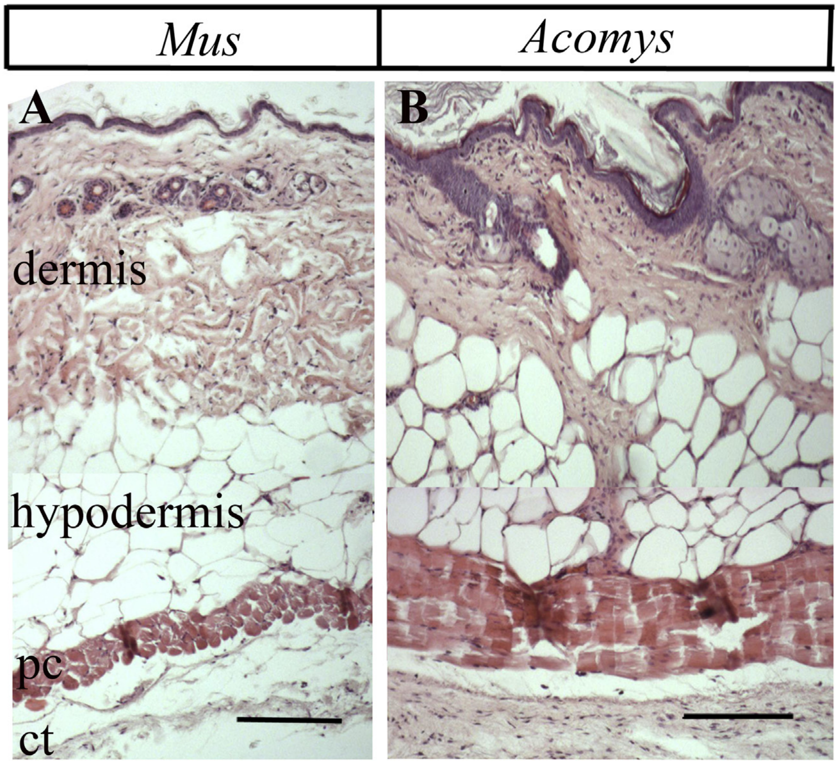

Fig. 1 – Normal structure of *Mus* and *Acomys* skin. A,***Mus* and** B, ***Acomys* shows the epidermis with hair follicles penetrating into the dermis, the collagenous dermis, the hypodermis with large adipocytes and the skeletal muscle of the panniculus carnosus (pc) and below that the fascial connective tissue (ct). In *Acomys*** (B) **there are also connective tissue spurs which penetrate the hypodermis and attach to the pc. Scale bars = 200 μm.**

in the third week it falls off because re-epithelialization of the injury site has by now taken place. The disappearance of the scab at 3 weeks reveals a smooth epidermis but the new dermal wound bed tissue has not yet completely filled in and there is often a gap in the center of the wound (Fig. 2C). The *Acomys* scab, by contrast comes away from the wound earlier and frequently in pieces because re-epithelialization is faster in *Acomys*. This can be seen in Fig. 2B where even as early as 1 week after injury the epithelium has in some places progressed half way to the center of the wound and displaced the scab in that area. By 3 weeks in *Acomys* (Fig. 2D), as is the case with *Mus*, the dermal part of the wound bed has not completely filled in and there is frequently a gap in the center.

Over the subsequent weeks of observation the *Mus* burn wound remains stable with a clear epithelium over the surface (Fig. 2E, G and I). In contrast in the 5 week *Acomys* burn small hairs can be seen to be emerging onto the surface of the wound which become very obvious at 6 weeks (Fig. 2H). These new spiny hairs seem to emerge first from the central region of the wound but over the next few weeks the other areas regenerate hairs so that the whole burn becomes covered with hairs, in this example at 8 weeks after injury (Fig. 2J).

3.3. *Histological aspects of burn injury*

The full thickness burns were sampled every 2/3 days for the first 2 weeks and then weekly. They were analyzed histologically, for cell proliferation with an antibody to PCNA and for the presence of various macrophage types by immunocytochemistry.

In both *Mus* and *Acomys* the result of the burn injury was to kill the epidermis and dermis which was reduced to a dense, hard layer and to de-cellularize the adipose layer and the skeletal muscle of the panniculus carnosus. The latter two layers remained in place and remained alive although they were shrunken (most apparent in Acomys, Fig. 3H) and showed much vascular leakage (Fig. 3A and H). The wound epithelium migrated across the burn through the 'dead' adipose layer and above the panniculus carnosus and after 3 days had progressed only a few hundred microns in both *Mus* and *Acomys* (Fig. 3D and K). There was a ring of thickened and proliferating epidermis 1 mm–1.5 mm wide around the edge of the injury region, the zone of hyperemia, in which there was no structural damage (Fig. 3A, H zh). In this zone, the epithelium was far thicker than the normal epidermis (normal *Mus* epidermis Fig. 3E, thick *Mus* epidermis 3F, thick *Acomys* epidermis Fig. 3J) and contained more proliferating cells which had been stimulated by the death of the adjacent burnt epithelium (Fig. 3F and J, PCNA stained epidermis). The migrating wound epithelium itself showed reduced levels of proliferation compared to the epithelium in the zone of hyperemia from which it arose (Fig. 3G vs F). Adjacent to this was the zone of stasis (Fig. 3A and H zs) which showed a huge inflammatory response and vascular leakage. In this region

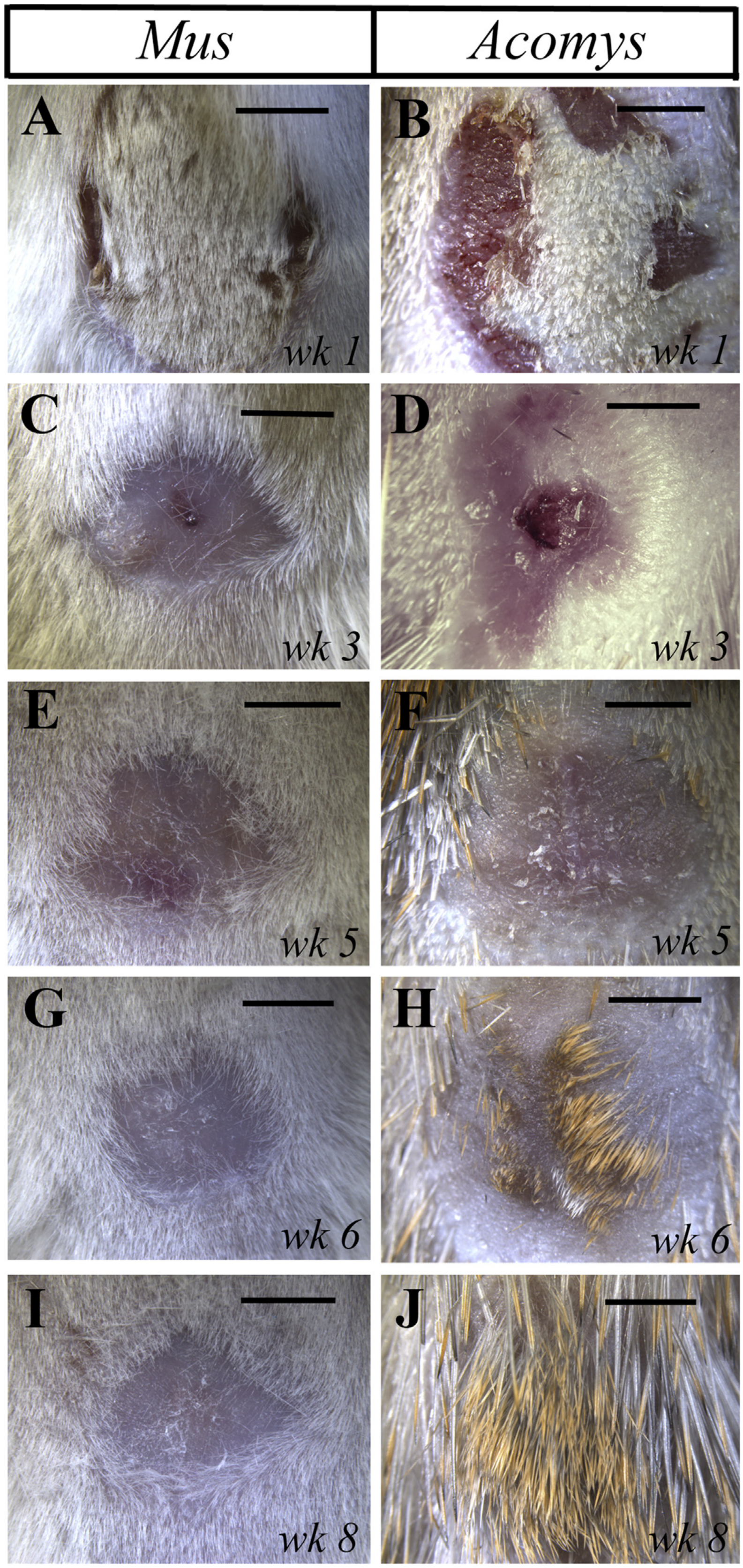

Fig. 2 – External views of *Mus* and *Acomys* dorsal skin in response to a full thickness burn injury. In week 1 the scab is still present (A, B) but it is clear that re-epithelialization is occurring faster in *Acomys* (B). By week 3 (C, D) the scabs have disappeared in both species to reveal incomplete filling in of the dermal wound bed. By week 5 the *Mus* wound is clear (E) but the *Acomys* wound begins to show the appearance of new hair follicles (F). By week 6 large golden spiny hairs are beginning to grow in *Acomys* (H). By 8 weeks this hair regeneration process has been completed in *Acomys* (J) whereas the Mus wound till remains hairless (I). Scale bars = 500 μm.

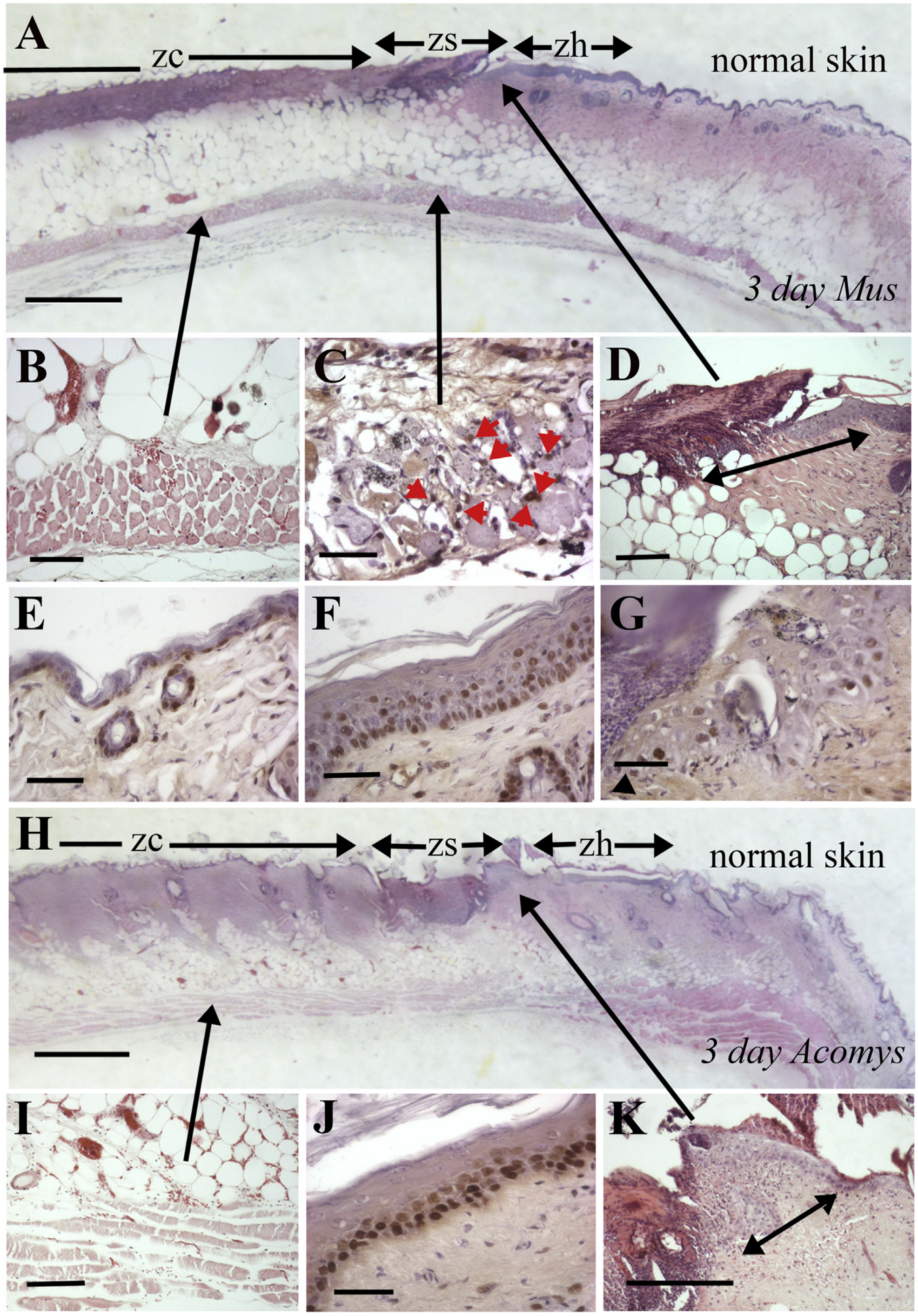

Fig. 3 – Mus and Acomys burn wound histology 3 days after injury. The zones of injury and the tissue specific responses are shown in *Mus* (A–G) and *Acomys* (H–K). A, overall image of *Mus* showing the transition from normal skin to zone of hyperemia (zh) to the zone of stasis (zs) to the zone of coagulation (zc). Scale bar = 500 μm. B, the panniculus carnosus in the zc region showing vascular leakage and shrunken fibers. C, the panniculus carnosus in the zs region which is beginning to regenerate and show proliferating cells (red arrows). D, the migrating wound epithelium with the distance travelled under the burnt dermis marked by the arrow. E, normal *Mus* skin section showing PCNA + ve cells in the two hair follicles and the basal layer of the epidermis. F, epithelium in the zh region showing a greatly thickened layer with more proliferating cells. G, the migrating epithelial wound front showing reduced numbers of proliferating cells compared to the non-migrating zone behind it (F). Arrowhead marks the migrating front of the epithelium. Scale bars in B–G = 50 μm. H, overall image of *Acomys* showing the transition from normal skin to zone of hyperemia (zh) to the zone of stasis (zs) to the zone of coagulation (zc). Scale bar = 500 μm. I, the panniculus carnosus in the zc region showing vascular leakage and shrunken fibers. J, epithelium in the zh region showing a greatly thickened layer with more proliferating cells. K, the migrating wound epithelium with the distance travelled under the burnt dermis marked by the arrow. Scale bars in I–K = 50 μm.

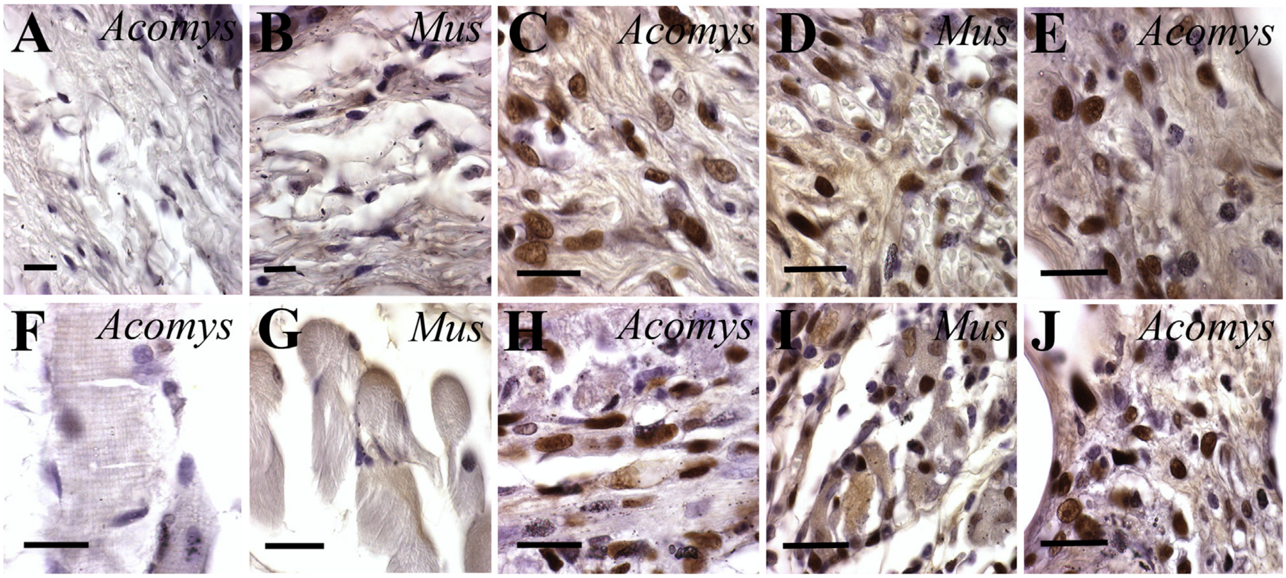

Fig. 4 – PCNA +ve (proliferating) cells in normal and injured tissues of *Mus* and *Acomys*. A–B, **no proliferating cells in the normal *Acomys*** (A) **or *Mus*** (B) **dermis.** C–D, **high levels of proliferating cells in the day 7 regenerating dermis of *Acomys*** (C) **and *Mus*** (D) **below the migrating epidermis.** E, **proliferating cells in a day 7 connective tissue spur in the *Acomys* hypodermis.** F–G, **normal skeletal muscle fibers in the panniculus carnosus showing no proliferating cells in *Acomys*** (F) **and *Mus*** (G). H–I, **regenerating pc muscle on day 7 showing many proliferating cells in both *Acomys*** (H) **and *Mus*** (I). J, **connective tissue around the adipose cells of the hypodermis in *Acomys* at day 7 showing many +ve cells. Scale bars = 50 μm.**

proliferating cells were seen in the adipose layer and in the panniculus carnosus suggesting that the skeletal muscle had begun regeneration (Fig. 3C). Internal to this region in the zone of coagulation where the epidermis and dermis were shrunken and dead, the adipose cells and skeletal muscle contained many red blood cells and the muscle fibers were shrunken in appearance (Fig. 3B and I). There were no obvious differences between these histological and proliferative events in *Mus* and *Acomys* at this early time point.

On days 5 and 7 the basic features of the burn injury described above changed little except for the continuing re-epithelialization which seemed to proceed at a faster rate in *Acomys* than in *Mus*. Most noticeable was the huge increase in proliferation reflected in the increased numbers of PCNA +ve cells in the dermis, the connective tissue around the adipocytes and in the regenerating zone of the panniculus carnosus in both species. Normal dermis shows almost no proliferative cells (0–5%) (Fig. 4A and B), but the dermis adjacent to the burn wound in the zone of hyperemia showed >50% PCNA +ve cells (Fig. 4C and D). Fibroblasts in the connective tissue strands in *Acomys* (Fig. 1B) also showed similarly high levels of proliferation (Fig. 4E). Likewise normal skeletal muscle of the panniculus carnosus shows almost no proliferative cells (Fig. 4F and G), but in the regenerative zone of the zone of stasis there were high levels (>50%) of proliferating cells, presumably satellite cells in both species (Fig. 4H and I). The tissue in between the dermis and the panniculus carnosus, the adipocytes of the hypodermis which are surrounded by connective tissue also showed high levels of proliferation (Fig. 4J). Thus viable cells throughout all the tissues of the skin adjacent to the burn edge begin proliferation after damage reaching similarly high levels by the end of the first week both in *Mus* and *Acomys*.

The second week after injury is characterized by continuingly more rapid re-epithelialization in *Acomys* than *Mus* and the appearance of hair placodes in the *Acomys* wound epithelium. Precise measurements of the rate of re-epithelialization *in vivo* are difficult due to the inhibitory presence of the scab. *In vitro* measurements using scratch assays have revealed that *Acomys* keratinocytes migrate twice as fast as *Mus* under these circumstances. The wound epithelium was noticeably thicker in *Mus* (Fig. 5A and E) than in *Acomys* (Fig. 5B and D) and in the *Acomys* wound epithelium newly developing hair follicles were apparent for the first time at the placode stage (arrow Fig. 5B). Hair placodes were not present in the *Mus* wound epithelium because no hairs are going to regenerate. In the dermis adjacent to these *Acomys* epithelial placodes there was a noticeable condensation of proliferating cells (arrow Fig. 5B) and there were more proliferating cells in this papillary layer of the *Acomys* dermis than in the reticular layer below (p vs r, Fig. 5B). No obvious papillary layer was apparent in the *Mus* regenerating dermis (Fig. 5A). Proliferating PCNA +ve cells were present in the regenerating dermis (Fig. 5A and B), hypodermis and panniculus carnosus of both species as these tissues continued to regenerate.

By 3 weeks the wounds had almost completed healing (1.5 cm wounds) but not entirely (Fig. 2C and D). There were large numbers of hair follicles in the *Acomys* wound epithelium at the germ and peg stages (Fig. 5C and inset) while the basal wound epithelial cells maintained high proliferation levels. In the newly formed papillary dermis below these follicles proliferating cells were still present but at lower density than

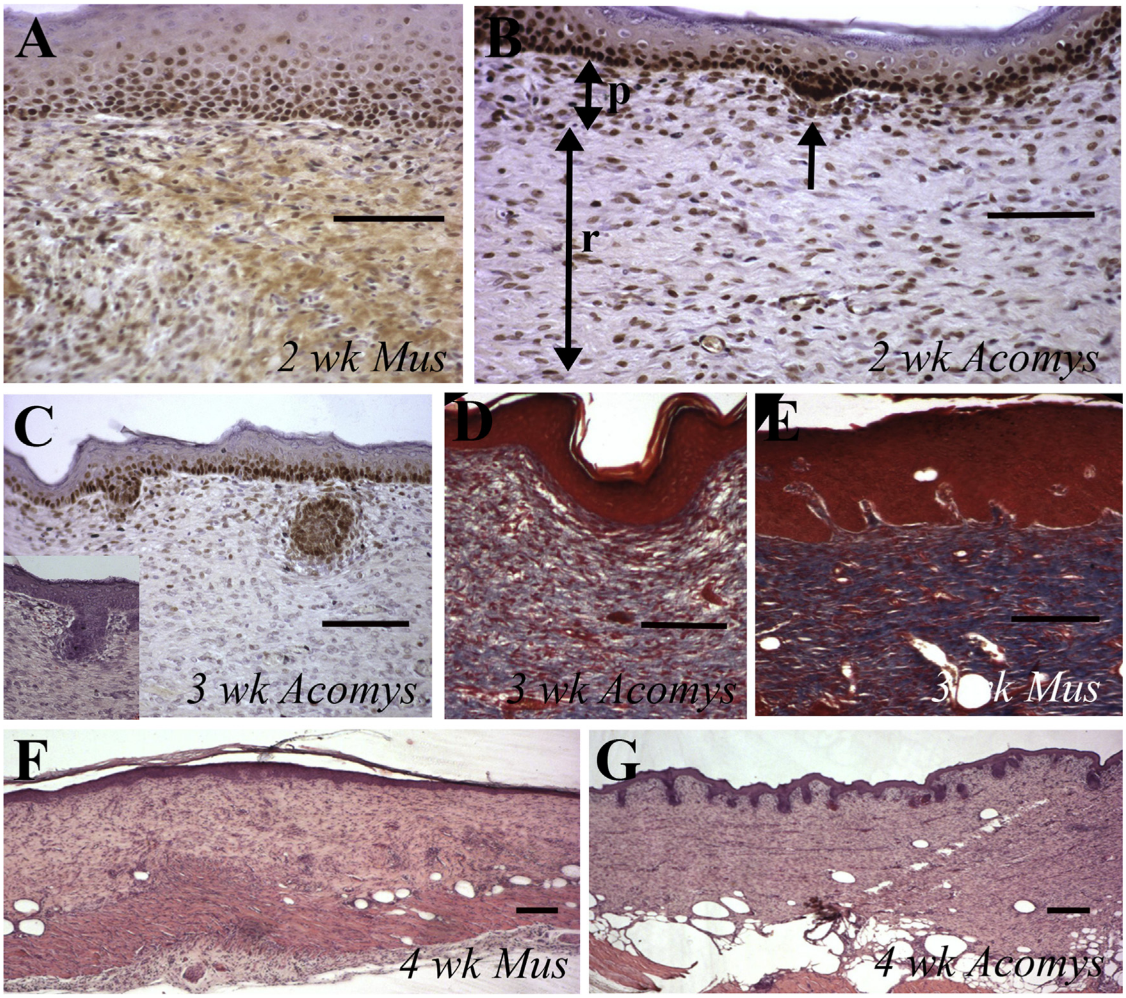

Fig. 5 – Regenerated tissues at 2–4 weeks after burn injury. A, **PCNA immunocytochemistry of *Mus* regenerated dermis below the wound epithelium at 2 weeks. The epithelium contains many proliferating cells, but is much thicker than the equivalent *Acomys* epithelium in** B**. Some dermal cells are PCNA +ve.** B, **PCNA immunocytochemistry of *Acomys* regenerated dermis and wound epithelium at 2 weeks. The epithelium is thinner than *Mus* (**A**) with proliferating cells in the basal layer and in a developing hair placode (arrow). There are high numbers of PCNA +ve cells in the dermis and the upper dense layer of cells in the papillary layer (p) including the condensation of cells below the hair placode (arrow). The lower, less dense layer is the reticular layer (r). Scale bars in A, B = 50 μm.** C, **PCNA immunocytochemistry of the regenerated wound in *Acomys* at 3 weeks after burn injury. The proliferating wound epithelium has two hair follicles developing, one at the hair bud stage and one at the hair germ stage. The insert shows a follicle at the peg stage.** D, **Trichrome staining of the regenerated epithelium and papillary layer of dermis 3 weeks after injury in *Acomys*. There is only a low level of blue staining suggesting low collagen levels.** E, **An equivalent region to** D **in *Mus* where the Trichrome staining show intense blue coloration in the upper dermis suggesting high levels of collagen in the matrix. The epidermis is also much thicker than in** D**. Scale bars in C – E = 100 μm.** F, **haematoxylin and eosin stained section of a 4 week regenerated skin in *Mus* showing a thickened epidermis, a uniform dermis and regenerating muscle in the pc below.** G, **haematoxylin and eosin stained section of a 4 week regenerated skin in *Acomys* showing many hair follicles developing from the epithelium and a more layered dermis. Scale bars in F, G = 100 μm.**

at 2 weeks (Fig. 5B vs C) and again, there seemed to be no clear papillary layer in the *Mus* dermis. In the lower dermal layer, the reticular dermis proliferation had virtually ceased in contrast to the same tissue at 2 weeks (Fig. 5B vs C). Trichrome staining of the *Acomys* dermis showed only low intensity of blue coloration suggesting low levels of collagen were present in the matrix (Fig. 5D). By contrast the *Mus* wound epithelium had no hair follicles and, as in the 2 weeks samples, was much thicker than the *Acomys* epithelium (Fig. 5E vs D). Below the *Mus* wound epithelium high levels of collagenous matrix staining was present after trichrome staining (Fig. 5E). Proliferation was still present in the basal layers of the wound epithelium whereas in the dermal layers of the *Mus* and *Acomys* wound proliferation had declined to low levels (Fig. 5C).

At 4–5 weeks the wounds had completed healing and the *Acomys* epithelium was full of hair follicles (Fig. 5G) some of which will become very large, forming spiny hair follicles (Fig. 6H). The basal layer of epithelial cells was still undergoing proliferation, as was the equivalent layer in *Mus* although there were no hair follicles present in *Mus* (Fig. 5F). Proliferation in the

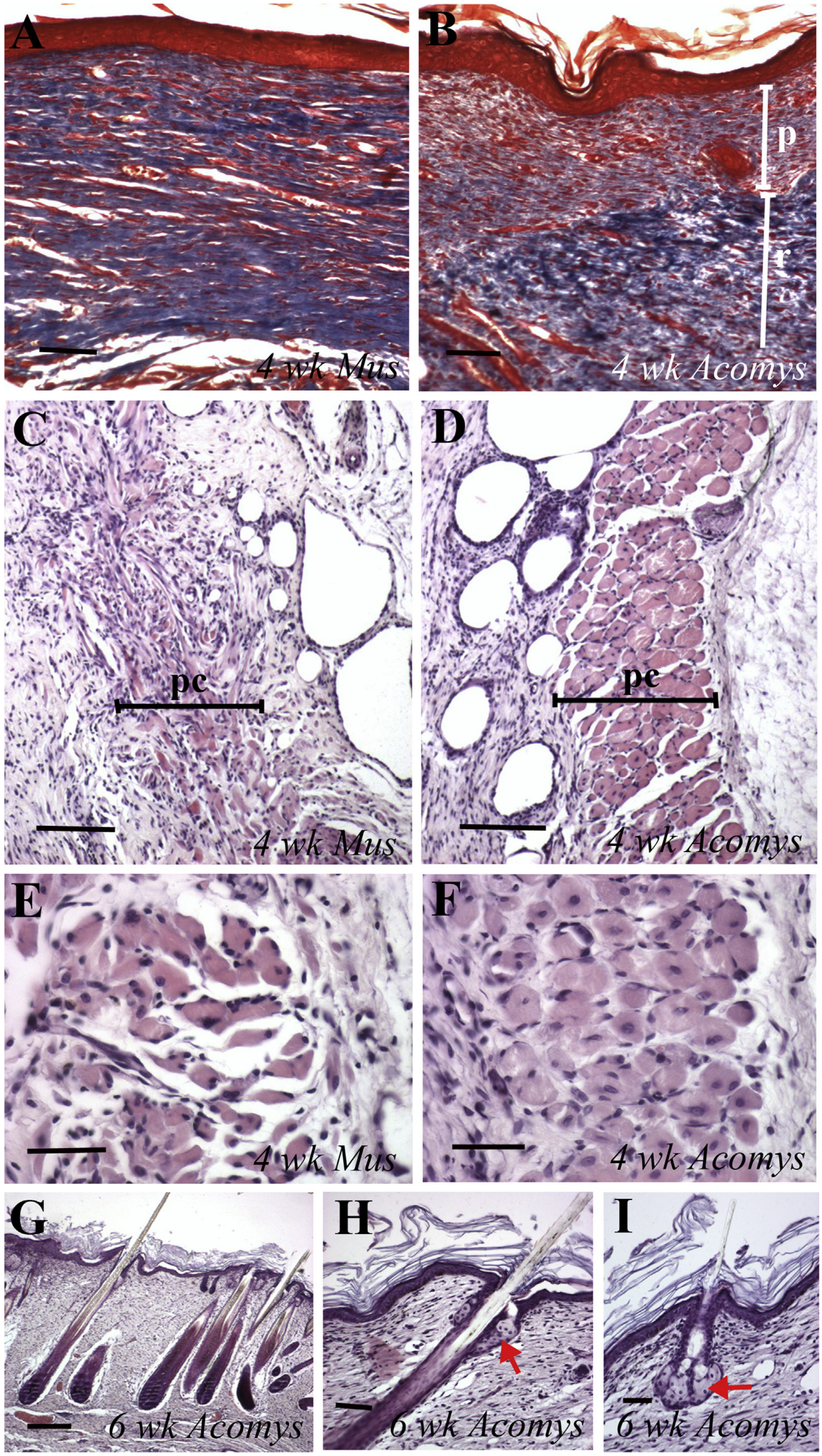

Fig. 6 – Regenerated tissues at 4-6 weeks after injury. A, **Trichrome stained dermis of *Mus* 4 weeks after burn injury showing a uniform blue, collagenous matrix.** B, **equivalent area of the *Acomys* dermis at 4 weeks showing a paler upper papillary layer (p) suggesting low matrix levels of collagen and the more intense collagen staining of the reticular layer (r) below.** C, **panniculus carnosus (pc) layer of the 4 week *Mus* burn wound showing regenerated fibers but in a very disorganized fashion.** D, **panniculus carnosus (pc) of the 4 week *Acomys* burn showing a perfectly regenerated stipe of skeletal muscle. Scale bars in A–D = 100 μm.** E,

dermal fibroblasts was now at a minimum and remained so for the rest of the observational period in both species. During this fourth week a clear distinction developed in the regenerated *Acomys* dermis between the papillary dermis underneath the epithelium and which was likely responsible for inducing the hair follicles and the reticular dermis in terms of collagenous matrix with trichrome staining. In *Acomys*, the papillary dermis was pale staining and formed a clear layer above the darker staining reticular dermis (Fig. 6B). In the regenerated *Mus* dermis, by contrast, there was no apparent light stained papillary dermis and a dark collagenous matrix was present throughout the dermal layer (Fig. 6A). By four weeks, regeneration of the skeletal muscle of the panniculus carnosus in *Acomys* had been completed throughout the wound and a new continuous layer of skeletal muscle was reformed (Fig. 6D) with many clearly regenerated fibers present with centrally located nuclei (Fig. 6F). In *Mus* although regeneration had taken place and fibers with centrally located nuclei were present (Fig. 6E) the muscle layer was often incomplete, twisted and fibrotic and by no means as perfect as that in *Acomys* (Fig. 6C).

There was little further change in the 6–9 week burns with little proliferation in the mesenchyme of both species and continuing basal cell layer proliferation in the epidermis of both species. There were virtually no adipose cells in the hypodermis which had regenerated in either species. In *Acomys* the large spiny hairs continued to elongate (Fig. 6G) and the smaller guard hairs also grew in between the spiny hairs. Both types were judged as functional since sebaceous glands were present (red arrows, Fig. 6H and I).

3.4. *Macrophage presence*

Using a panel of 5 different macrophage markers to identify M1 pro-inflammatory macrophages (CD86, IBA-1 and F4/80), M2 anti-inflammatory macrophages (CD206) or tissue macrophage/monocytes (MOMA-2) we asked whether there was a differential distribution between these sub-types in *Mus* vs *Acomys* or between the individual components of the burnt skin over the first 2 weeks after injury.

In general, macrophages rapidly appeared throughout the connective tissue layer of the skin in the zones of hyperemia and stasis adjacent to the edge of the burn. They then seemed to spread upwards into the muscle of the panniculus carnosus and then around the adipose cells of the hypodermis. The dermis and epidermis were not invaded by macrophages and formed a scab which was eventually shed from the skin. There were significant differences between the two species in terms of macrophage subtype presence or absence.

MOMA-2 +ve cells were present uniformly throughout the three tissue layers of *Mus* by day 7 as seen in Fig. 7A (FITC fluorescence=MOMA-2, DAPI=nuclei). No MOMA-2 +ve cells could be detected in corresponding sections of *Acomys* skin (Fig. 7B) as reflected in the absence of FITC fluorescence in the section, but many nuclei. Only a pale green background can be seen in the pc muscle tissue. Using the F4/80 macrophage marker on *Mus* sections positive cells were primarily concentrated in the connective tissue layers below the pc muscle on day 7 (Fig. 7C) where strong FITC fluorescence was seen. In contrast, no F4/80 positive macrophages were seen in sections of *Acomys* burns at any stage (Fig. 7D) and in this image only a pale background staining of the muscle can be seen despite the presence of many DAPI stained nuclei. This is a similar result to the lack of F4/80 +ve macrophages in *Acomys* excisional skin wounds despite their presence in *Acomys* spleen and the induction of markers of M1 macrophages such as iNos and IFN-γ by activated spleen cells [6].

Two other markers of M1 inflammatory macrophages were also used, namely IBA-1 and CD86. In the case of IBA-1 many +ve macrophages were seen in both *Mus* (Fig. 7E) and *Acomys* (Fig. 7F) widely spread throughout the three tissue layers in both species and were also present at high levels in the adjacent living dermis. CD86 +ve cells were present both in *Mus* (Fig. 7G) and *Acomys* (Fig. 7H) sections and they seemed mostly concentrated in the connective tissue below the panniculus carnosus with considerably fewer cells appearing in the muscle of the pc. Finally CD206 was used as a marker of the M2 anti-inflammatory phenotype. This gave the most intense immunoreactivity of all the antibody markers and was present strongly throughout all the tissue layers of both *Mus* (Fig. 7I and *Acomys* (Fig. 7J) damaged tissues.

4. Discussion

The inevitable result of deep thermal injury to adult mammalian skin is a permanent scar, disfigurement and loss of mobility if the injury traverses a joint. Because there is no endogenous regenerative capacity of the skin current treatments are restricted to skin grafting and the use of tissue engineered dermal substitutes to improve outcomes [7]. Interestingly adipose tissue seems to possess some regeneration inducing capacities and it has been used to improve skin texture [8], increase vascularity, VEGF and SDF-1 levels and decrease MMP8 and collagen deposition after grafting into thermal burn sites [9]. Using purified ASCs also improved vascularity and collagen III levels in burn wounds [10] and there was also some suggestion that additional hair follicles were produced in these experiments.

Regardless of whether ASCs were administered or not (but they are extensively present in the hypodermis) we show here that *Acomys*, the spiny mouse, can perfectly regenerate after a

Higher magnification view of the 4 week *Mus* pc showing individual regenerated fibers with centrally located nuclei but the fibers seems twisted and disorganized. F**, pc of the 4 week *Acomys* wound showing well organized regenerated fibers. Scale bars in E–F = 50 μm** G**, large spiny hairs in the 6 week *Acomys* burn wound.** H**, higher magnification of a 6 week regenerated spiny hair from *Acomys* showing the sebaceous glands (red arrow) just below the epidermis.** I**, higher magnification of a 6 week regenerated guard hair in the *Acomys* regenerated wound showing sebaceous glands (red arrow) present. Scale bars in G–I = 100 μm.**

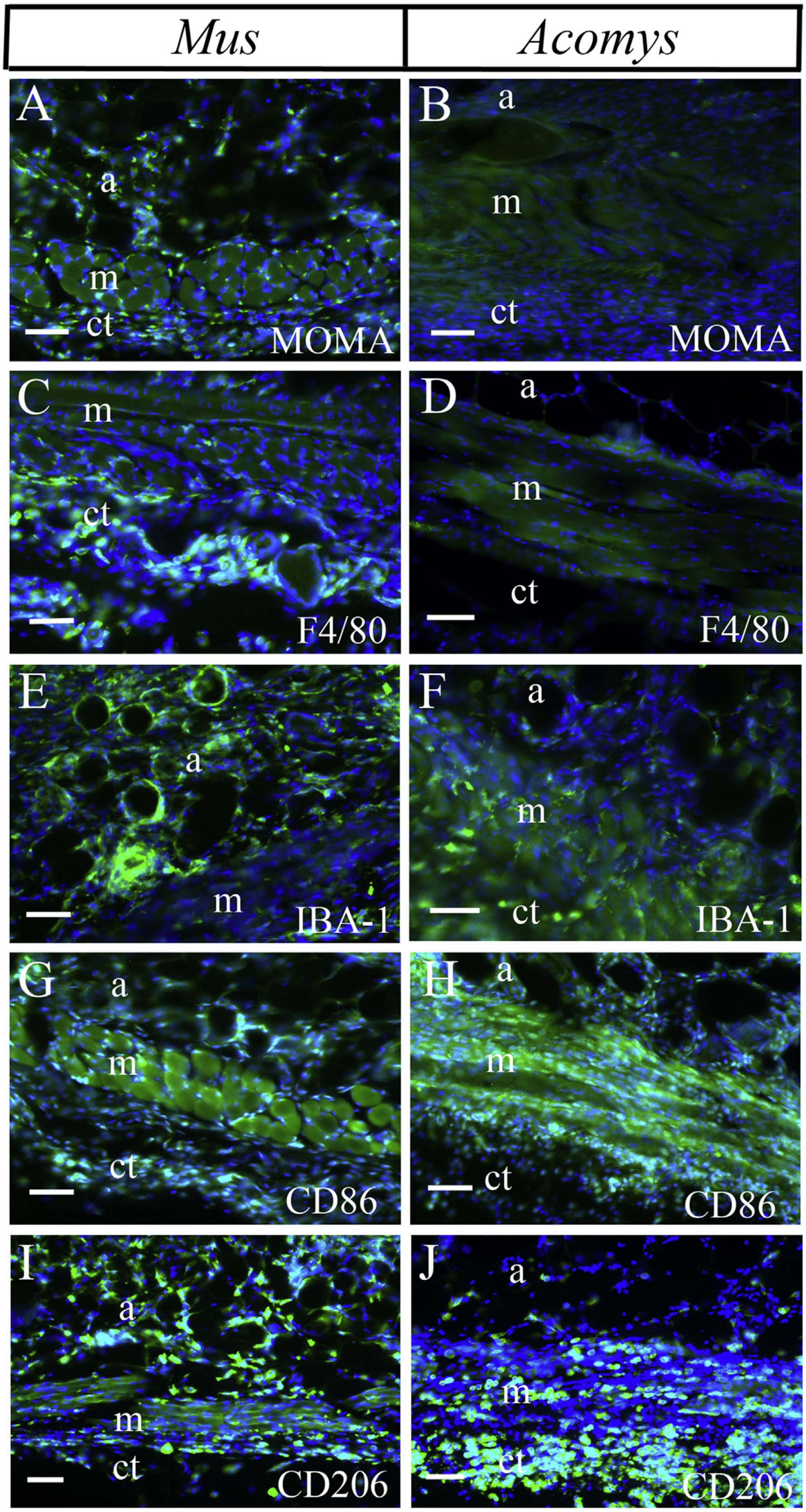

Fig. 7 – Macrophages in injured tissues. Macrophage antibody immunoreactivity on sections of *Mus* (A, C, E, G, I) and *Acomys* (B, D, F, H, J) 7 days wounds showing the presence or absence of different subtypes. A, B**=MOMA-2.** C, D**=F4/80.** E, F**=IBA-1.** G, H**=CD86.** I, J**=CD206. a=adipose cells in the hypodermis; m=skeletal muscle of the panniculus carnosus; ct=connective tissue of the lowest layer. Scale bars=100 μm.**

relatively large burn injury to the dorsum. After the formation of a scab which is the size of the burn injury (1.5 cm diameter) and consists of dead epidermal and dermal cells, the epidermis grows under the scab through the hypodermal adipose cell layer. After completion of re-epithelialization the scab is released from the wound. The non-regenerating *Mus* epithelium also grows under the scab and ejects it from the skin surface, but the rate of re-epithelialization is slower (about half the rate) in *Mus*. As re-epithelialization takes place the dermis re-appears by migration into the wound site from the undamaged edges and high levels of proliferation take place in these dermal fibroblasts. The same phenomenon occurs in the *Mus* wound, re-epithelialization and dermal fibroblast proliferation, and the two wounds look quite similar except for a thicker epithelium in *Mus* and the complete absence of new hair follicles which in *Acomys* extend all across the new wound epithelium. After the formation of these new hair follicles in *Acomys* the upper layer of dermis, the papillary layer, does not produce a dense collagenous matrix as the *Mus* dermal fibroblasts do which likely reflects intrinsic differences in the properties of these fibroblasts not only in their secretory capacities but also in their signaling capacities to induce the formation of new hair follicles.

The skeletal muscle layer of the panniculus carnosus also shows important differences between the two species in the response to thermal injury. The connective tissue structure of the muscle was not removed by the injury and only the myofibers themselves were damaged. This can be seen in the shrunken look that the muscle took on with the maintenance of the endomysium and perimysium as it becomes invaded with macrophages. It might be expected that the *Mus* panniculus carnosus would regenerate well since the connective tissue plays an important role in muscle regeneration. Skeletal muscle in mammals does regenerate after injury such as cardiotoxin administration which causes a breakdown of the myofiber plasma membrane in contrast to volumetric muscle loss where a section of the muscle is completely removed and no regeneration ensues [11,12]. Even so, the *Mus* panniculus carnosus did replace muscle fibers but in a disorganized and tangled manner and not completely throughout the wound as there was a good deal of fibrosis present. This suggests that the thermal injury had done some damage to the molecular components of the endomysial connective tissue to result in defective regeneration. In contrast the *Acomys* panniculus carnosus regenerated perfectly despite a similar level of thermal damage suggesting that *Acomys* muscle regeneration may not depend upon the preservation the connective tissue. Indeed, both histological [6] and our unpublished molecular analyses of skin regeneration after full thickness excision has revealed that the panniculus carnosus can regenerate the hole that has been created following skin removal.

All the surviving tissues of the skin became infused with macrophages after thermal wounding. Since macrophages play an important role in muscle regeneration [13] it is likely that the same applies here in burn injury because muscle was a large component of the regenerated tissue. It has been shown that MI pro-inflammatory macrophages are involved in the clearance of necrotic debris and M2 pro-regenerative macrophages provide growth factors to stimulate myofiber regeneration [13,14]. This situation would certainly apply here to these burn injuries because there were large numbers of M1 macrophages identified with the IBA-1 and CD86 markers and M2 macrophages identified by the CD206 marker throughout the hypodermis, muscle and connective tissues in both species. However there was an absence of both MOMA-2 and F4/80 macrophages in *Acomys* suggestive of a reduced inflammatory response. Since the F4/80 macrophage phenotype can be induced in *Acomys* bone marrow cells by IFN-γ and LPS [15] it imples that the cytokine environment of the *Acomys* injured skin is different from that of *Mus* as we have observed following excisional skin wounds [6]. The further identification and characterization of these differences will be important and may be, at least partially, responsible for the very different regenerative responses of *Mus* and *Acomys* tissue to injury.

Declaration of interests

None.

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REFERENCES

[1] Finnerty CC, Jeschke MG, Branski LK, Barret JP, Dziewulski P, Herndon DN. Hypertrophic scarring: the greatest unmet challenge after burn injury. Lancet 2016;388:1427–36.
[2] Amini-Nik S, Yousuf Y, Jeschke MG. Scar management in burn injuries using drug delivery and molecular signaling: current treatments and future directions. Adv Drug Deliv Rev 2018; 123:135–74.
[3] Tredget EE, Levi B, Donelan MB. Biology and principles of scar management and burn reconstruction. Surg Clin North Am 2014;94:793–815.
[4] Seifert AW, Kiama SG, Seifert MG, Goheen JR, Palmer TM, Maden M. Skin shedding and tissue regeneration in African spiny mice (Acomys). Nature 2012;489:561–5.
[5] Brant JO, Lopez M-C, Baker HV, Barbazuk WB, Maden M. A comparative analysis of gene expression profiles during skin regeneration in Mus and Acomys. PLoS One 2015;25:e0142931.
[6] Brant JO, Yoon JH, Polvadore T, Barbazuk B, Maden M. Cellular events during scar-free healing in the spiny mouse, Acomys. Wound Rep Regen 2016;24:75–88.
[7] van Zuijlen PPM, Gardien KLM, Jaspers MEH, Bos EJ, Baas DC, van Trier AJM, et al. Tissue engineering in burn scar reconstruction. Burns Trauma 2015;3:18.
[8] Klinger M, Marazzi M, Vigo D, Torre M. Fat injection for cases of severe burn outcomes: a new perspective of scar remodeling and reduction. Aesthetic Plast Surg 2008;32:465–9.

[9] Sultan SM, Barr JS, Butala P, Davidson EH, Weinstein AL, Knobel D, et al. Fat grafting accelerates revascularisation and decreases fibrosis following thermal injury. JPRAS 2012;65: 219–27.
[10] Bliley JM, Argenta A, Satish L, McLaughlan MM, Dees A, Tompkins-Rhoades C, et al. Administration of adipose-derived stem cells enhances vascularity, induced collagen deposition, and dermal adipogenesis in burn wounds. Burns 2016;42:1212–22.
[11] Charge SBP, Rudnicki MA. Cellular and molecular regulation of muscle regeneration. Physiol Rev 2004;84:200–38.
[12] Musaro A. The basis of muscle regeneration. Adv Biol 2014 Article ID 612471.
[13] Sunman M, Warren GL, Mercer RR, Chapman R, Hulderman T, van Rooijen N, et al. Macrophages and skeletal muscle regeneration: a clodronate-containing liposome depletion study. Am J Physiol Regul Integr Comp Physiol 2006;290: R1488–95.
[14] Heredia JE, Mukundan L, Chen FM, Mueller AA, Deo RC, Locksley RM, et al. Type 2 innate signals stimulate fibro/ adipogenic progenitors to facilitate muscle regeneration. Cell 2013;153:376–88.
[15] Simkin J, Gawriluk TR, Gensel JC, Seifert AW. Macrophages are necessary for epimorphic regeneration in African spiny mice. eLife 2017;6:e24623.